ITA0611 - MACHINE LEARNING

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Name: M. Shirija
Reg: 192412511
Slot: D
Faculty: Dr. K Sivakumar

A factory has three boxes containing coloured balls. Box A contains 6 red, 4 blue, and 2 green balls. Box B contains 5 red, 1 blue and 3 green balls. One box is selected at random, but box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Sol: Probability that the selected box was box C

Given that

Box A: 6 Red, 4 blue, 2 Green → Total = 12

Box B: 5 Red, 1 blue, 3 Green → Total = 15

Box C : 3 Red, 2 blue, 5 Green → Total = 17

Box B is twice as likely as A and C

$$P(A) = \frac{1}{4} \; ; \; P(B) = \frac{1}{2} \; ; \; P(C) = \frac{1}{4}$$

Probability of drawing blue:

$$P(\text{Blue}|A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue}|B) = \frac{1}{15}$$

$$P(\text{Blue}|C) = \frac{2}{15}$$

Using Bayes theorem:

$$P(C|\text{Blue}) = \frac{P(C)P(\text{Blue}|C)}{P(A)P(\text{Blue}|A) + P(B)P(\text{Blue}|B) + P(C)P(\text{Blue}|C)}$$

$$= \frac{\frac{1}{11} \times \frac{2}{15}}{\frac{1}{11} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{11} \times \frac{2}{15}}$$

$$= \frac{1/30}{1/20} = \frac{2}{21}$$

$$\therefore P(C|\text{Blue}) = \frac{2}{21} = 0.0952$$

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; section B has 4 red, 6 blue and 5 black cars; and section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from section C.

Given

Section A: 5R, 3B, 2Black → Total = 10

Section B: 4R, 6B, 5Black → Total = 15

Section C: 7R, 2B, 6Black → Total = 15

Section B is twice as likely:

$$P(A) = \frac{1}{4}; \quad P(B) = \frac{1}{2}; \quad P(C) = \frac{1}{4}$$

Blue probabilities:

$$P(\text{Blue}|A) = 3/10$$

$$P(\text{Blue}|B) = 6/15 = 2/5$$

$$P(\text{Blue}|C) = 2/15$$

ing Bayes theorem:

$$P(C|Blue) = \frac{1/4 \times 2/5}{1/4 \times 3/10 + 1/2 * 2/5 + 1/4 \times 2/5}$$

$$= \frac{1/30}{37/120} = 4/37$$

$$\therefore P(C|Blue) = 4/37 = 0.1081$$

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with K-Nearest neighbours algorithm is applied for binary classification. The dataset contains of 1800 samples per class, with 20% of the data reserved for testing. During evaluation the model reports 210 true +ves (TP), 190 true -ves (TN), and 30 FP and based on the given information, determine the missing false negatives value and then calculate the accuracy, precision, sensitivity and specificity of the classifier, explaining its effectiveness in medical diagnosis?

The dataset has:

- 1800 malignant
- 1800 benign

Total = 3600

20% is used for testing:

$3600 \times 20\% = 720$

given; TP = 210; TN = 190

Finding FN

$$FN = 420 (-) (TP + TN + FP)$$

$$= 420 - (210 + 190 + 30)$$

$$FN = 290$$

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{210 + 190}{420}$$

$$= \frac{400}{420}$$

$$\text{Accuracy} = 55.55\%$$

Precision:

$$\text{Precision} = \frac{TP + TN}{TP + FP + FN + TP}$$

$$= \frac{210 + 190}{210 + 30 + 210 + 190}$$

$$= \frac{400}{420}$$

$$\text{Precision} = 87\%$$

Sensitivity / Recall

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{210}{210 + 290} = \frac{210}{500}$$

$$\text{Sensitivity} = 42\%$$

ficity:

$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$= \frac{190}{190+30}$$

Specificity = 86.36%

4. Examine the performance of a ML-model using its training and validation loss values over multiple epochs

a. Epoch 1: Training loss = 2.8, validation loss = 2.41

b. Epoch 2: Training loss = 2.5, validation loss = 2.2

c. Epoch 3: Training loss = 2.2, validation loss = 2.0

d. Epoch 4: Training loss = 2.0, validation loss = 2.1

e. Epoch 5: Training loss = 1.8, validation loss = 2.3

f. Epoch 6: Training loss = 1.6, validation loss = 2.6

g. Epoch 7: Training loss = 1.5, validation loss = 3.2

h. Epoch 8: Training loss = 1.3, validation loss = 3.2

i. Identify the epoch at which the model begins to overfit

ii. Suggest two effective techniques to reduce overfitting in this model.

iii. Discuss how overfitting affects the model's generalization ability on unseen data

Sol: (i) Epoch at which overfitting begins

At epoch 3, validation loss is at its minimum

validation loss = 2.0

From epoch 4 onwards, training loss continues to decrease, but validation loss starts increasing.

(ii) Two techniques to reduce overfitting

1. Dropout: Randomly disables some neurons during training, reducing dependence on particular features.
2. Early stopping: Stops training when validation loss starts increasing, preventing and model from learning the training data too closely.

(iii) Effect of overfitting on generalization

overfitting causes the model to perform very well on training data but poorly on unseen data. Therefore, its generalization ability decreases.

5. In shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in shop 4, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with shop 4 being twice as likely to be chosen as shop X. The selected bottle is found to be adulterated.

The question states:

Shop X: 20 pure, 30 adulterated → Total = 50

Shop 4: 40 pure, 20 adulterated → Total = 60

Shop 4 is twice as likely to be selected as shop X

The selected bottle is adulterated.

Step 1: Prior probabilities

Since 4 is twice as likely:

$$P(X) = \frac{1}{3} \; ; \; P(4) = \frac{2}{3}$$

Step 2: Probability of adulterated bottle

$$P(A|X) = \frac{30}{50} = \frac{3}{5}$$

$$P(A|4) = \frac{20}{60} = \frac{1}{3}$$

Apply Bayes theorem

$$P(Y|A) = \frac{P(Y)P(A|Y)}{P(X)P(A|X) + P(Y)P(A|Y)}$$

$$= \frac{\frac{2}{3} \times \frac{1}{3}}{\frac{1}{3} \times \frac{3}{5} + \frac{2}{3} \times \frac{1}{3}}$$

$$= \frac{2/9}{1/5 + 2/9}$$

$$= 10/19$$

$$P(Y|A) = 10/19 = 0.5263$$